Longer by how much?

The note inside. · Curiosity day

Name _____ Date _____



A comparison model, not two measured piano strings. Compare length while imagining the material, thickness, and tension stay the same.

PRETEND DRAWING LENGTHS · CENTIMETERS

Line	A	B
Length (cm)	24	9

These are invented drawing lengths. Do not measure the painting or real piano strings.

1. Line A is 24 cm long. Line B is 9 cm long. How much longer is A?

24 − 9 = _____ Show your thinking.

2. Imagine drawing A and B end to end. What is their total length?

24 + 9 = _____ Show your thinking.

3. Little erases 6 cm from line A. How long is that line now?

24 − 6 = _____ Show your thinking.

Sketch two bars that start at the same place. Show which part is the extra 15 cm. A sketch need not be actual size.

WONDER DAILY

The note inside.

One small press

One key sinks beneath Little's hand. A note rings through the museum. "Was it hiding in there?" Little asks. Big listens beside the bench.

A hidden tap

An enormous model gives them a peek. A key tips; connected parts lift a felt-covered hammer. Tap! The hammer strikes a string and comes away.

After the tap

The hammer has fallen back, but the string is not still. It quivers: back and forth, back and forth. That quick motion is vibration.

The wood joins in

Little follows the same string to a wooden bridge. The bridge carries vibration into a broad soundboard. That wood vibrates too, moving more air.

Two lengths to notice

Two strings stretch across another model. One is longer. If everything else stays the same, the longer string vibrates more slowly and sounds lower.

Keep the path

With the pedal up, Little releases a middle key. A felt damper hushes its strings. Little sketches key, hammer, strings, wood.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

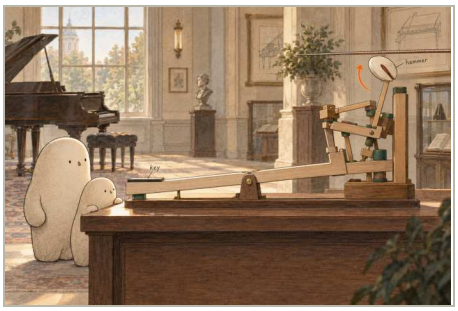
4 Fold into a book. Page 1 goes on the front!

Give the note a path.

The note inside. · Curiosity day

Name _____ Date _____

Choose one picture from Little's search. Tell what moves and what happens next. Draw or write a clue that helped you understand the note.



A hidden tap



The wood joins in



Two lengths to notice

Draw one detail from your scene. Add an arrow or a label to show a movement or a difference.

key · hammer · string · vibration · bridge · soundboard · pitch

When a key moves, … What does the action help the hammer do?

The soundboard helps because … How does the broad wood help move air?

If only the string's length changes, … Which length could make the lower sound?

Read it together and point to the detail your words explain. If you are writing, check the first capital and the ending mark.

The note inside.

01 Where could a note be hiding?



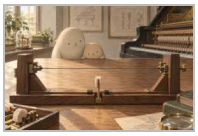
A tiny movement begins a question. This imagined museum piano is an acoustic piano, with strings inside.

Look closer. Find Little's hand and the orange arrow. One key moves only a little. Where would you look for what happens next?

A little explanation. The piano is not a box of stored notes. Pressing a key starts moving parts inside. In an acoustic piano, those parts send a small hammer toward metal strings. The next picture lets us look at a model of that hidden motion.

Try this. Pretend the tabletop is a piano key. Lower one finger quietly, then lift it. Say what you imagine moving inside.

02 How can a string keep singing after one tap?



One string, shown with exaggerated motion traces. Real piano-string vibrations are much smaller and faster than this drawing can show.

Look closer. Find the gap between the hammer and the string. Look for the two faint orange traces. What might they help us imagine?

A little explanation. A hammer taps the string and falls away. The string keeps moving back and forth very quickly. That motion is vibration. The orange traces exaggerate the movement; they are not extra strings. The vibration gradually fades unless something stops it sooner.

Try this. Move one finger a tiny distance back and forth in the air. Let it become still. Say when your pretend string could make sound.

03 How does a thin string fill a room with sound?



The bridge carries vibration into the soundboard. This cutaway leaves out parts so we can notice the connection.

Look closer. Find the narrow raised bridge and the broad wooden board. Where do the strings meet the bridge? Follow the orange mark into the wood.

A little explanation. A thin string alone moves little air. Its vibration passes through the bridge into the soundboard. This broad wooden board vibrates too, moving more air. The moving air carries sound to our ears. The wood helps us hear a small string's motion.

Try this. Trace the path with your finger: string, bridge, soundboard. Finish by pointing toward your ear. What helped the motion reach you?

04 Why do the two ends of a piano sound different?



A comparison model, not two measured piano strings. Compare length while imagining the material, thickness, and tension stay the same.

Look closer. The display has two strings. Trace each from one support to the other. What is different about the distance between the supports?

A little explanation. Pitch means how high or low a note sounds. With other things the same, a longer string gives a lower pitch and a shorter one gives a higher pitch. Tightening the same string raises its pitch; loosening it lowers the pitch. In a piano, length, thickness, and tension work together. Low notes are toward the player's left; high notes are toward the right.

Try this. Use your voice to make a comfortable low hum, then a higher hum. Keep both gentle. Can a high sound be quiet too?